

Designing Sustainable Cities

How smart ideas like Sponge Cities, Green Roofs, and Clean Mobility create healthier places to live.



Clean Nature



Smart Water



Green Power

What Makes a City Sustainable?



Zero Trash Goal

Reusing materials and recycling everything possible.



Space for Plants & Animals

Parks and green corridors brought directly into neighborhoods.



Sun & Wind Energy

Powering electricity using clean, renewable sources.



What is a Sponge City?

A **sponge city** is a city designed to **absorb, store, clean and slowly release rainwater**, instead of allowing rainwater to quickly run off into streets and cause flooding.

It works **like a sponge**—it soaks up water when there is heavy rainfall and can use or release the water later.



1. Absorb

Special pavements soak up rain instantly to prevent floods.



2. Store

Underground tanks hold water safely during heavy storms.



3. Clean

Plant roots filter dirty water naturally without chemicals.



4. Reuse

Cleaned rainwater waters park trees and fills city lakes.

Why Sponge Cities Work



80%

Rainwater Saved

Stops streets from flooding during heavy downpours.

2.5°C

Cooler Cities

Water ponds and grass cool down hot summer air.

Green Belts: Nature's Protective Ring

A **green belt** is an area of **protected open land, forests, farms, parks or vegetation around or within a settlement** where construction is restricted.

Stops City Sprawl

A protected ring of trees stops cities from swallowing farmland.

Homes for Wildlife

Gives birds and forest animals safe pathways to live and roam.

Giant Air Filter

Millions of trees soak up carbon dioxide and give us fresh oxygen.

Green Buildings: Smart Architecture

A **green building** is a building designed and constructed to **use fewer natural resources and cause less environmental damage**



Natural Sunshine

Big windows catch natural sunlight to save daytime electricity.



Eco Materials

Built with timber, bamboo, and recycled glass instead of heavy concrete.



Smart Sensors

Automatic lights turn off when a room is empty.

Green Rooftops & Living Walls

A **green roof** is a building roof that is partly or completely covered with **plants, soil or another growing medium**.



Keeps Buildings Cool

Rooftop plants act like a blanket that keeps heat out in summer.



Traps Rainwater

Soaks up half of the storm rain before it reaches the ground.



Rooftop Vegetables

Provides fresh local food and habitats for honeybees.

15–Minute Cities: Everything Nearby



Walk Everywhere

Schools, parks, and shops are just a 15-minute walk away.



Electric Buses

Quiet, pollution-free electric buses carry people fast.



Safe Bike Lanes

Protected cycle paths separated safely from big cars.

Zero Waste Goals

Circular Economy & Clean Energy

100%

Solar Powered

Zero

Landfill Waste

90%

Water Recycled



Trash to Power: Food scraps and organic waste are composted into natural fertilizer and biogas energy for homes.

Old Concrete City vs. Future Eco-City

× Old Concrete City

- Roads flood easily during big rainstorms
- Too many gas-powered traffic jams
- Massive piles of trash going to landfills
- Very hot in summer with few trees

✓ Future Eco-City

- + Sponge pavements soak up water safely
- + Walkable streets and electric buses
- + Everything is reused or recycled
- + Shaded parks and cool rooftop gardens

Step-by-Step

How Cities Make the Change

Step 01

Plan

Map out flood zones and hot concrete areas.

Step 02

Plant

Protect green belts and add rooftop plants.

Step 03

Pave

Replace hard concrete with sponge ground.

Step 04

Power

Install solar panels and electric bus lines.



Building Greener Cities for Our Future!

By combining nature with smart engineering, we build healthy, happy cities for everyone.

[#SpongeCities](#)

[#GreenRoofs](#)

[#EcoCities](#)

Image Sources



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Source: ecotourism-world.com



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